

Tripzy.travel: Autonomous Agentic Travel Recommendation & Viral Discount Engine

A comprehensive whitepaper detailing the core platform vision, cold-start recommendation algorithm, 3-layer architecture, frictionless hotel barter model, Google AdMob rewarded ads, and viral Instagram Flash Lotteries scaling to 100,000+ MAU.

Document Version:	3.0.0 (Master Whitepaper)	Target Market:	Turkey Launch & Global Expansion
Core Problem Solved:	Cold-Start & High Travel CAC (\$0 CAC)	Social Hub:	@tripzydeal (Instagram)
Tech Stack:	Next.js, FastAPI, Supabase, pgvector	Infrastructure Cost:	< \$400 / mo (>90% Net Margin)
Scale Target:	100,000+ Monthly Active Users	Regulatory Class:	MP■ Loyalty & KVKK Compliant

EXECUTIVE OVERVIEW & PLATFORM THESIS

Tripzy.travel is a next-generation travel discount platform engineered to overcome the two largest hurdles in travel tech: (1) the **Cold-Start Problem** (the inability of algorithms to personalize for users with zero booking history), and (2) **Extorbidant Paid Ad Acquisition Costs**.

By capturing high-frequency lifestyle micro-signals (Instagram Stories, double-taps, daily streaks, geofenced proximity) and projecting them into a 32-dimensional latent vector space, Tripzy provides instant personalization for brand-new users. Furthermore, Tripzy turns hotel off-peak barter inventory into viral Instagram Flash Lotteries, acquiring tens of thousands of users at zero cash CAC and monetizing non-winners through automated 48-hour consolation vouchers, Google Rewarded Ads, and VIP subscriptions.

1. THE INDUSTRY PROBLEM & 3-LAYER ARCHITECTURE

1.1 The Travel Cold-Start & High CAC Dilemma

Consumers purchase travel only 1–2 times per year, causing user-item interaction matrices to be over 99.9% empty. Standard travel apps show generic static lists and spend \$30–\$120 on ads to acquire a single booking. Meanwhile, boutique hotels and resorts face perishable unsold room inventory that produces zero revenue.

1.2 The Tripzy 3-Layer Solution

Architectural Layer	Key Technologies	Core Function & Responsibility
Layer 1: Interface & Signal Ingestion	Next.js PWA / React Native, Tailwind CSS, LocalStorage	Captures implicit & explicit micro-actions (double-taps, story dwell time, scratch card reveals, geofence watch). Buffers telemetry.
Layer 2: Autonomous Reasoning Brain	FastAPI (Python), Google Gemini LLMs, Pydantic, SVD++	Cross-Domain Lifestyle Projection (translates daily coffee/dining habits into travel preferences), SGD matrix factorization, provably fair draws.
Layer 3: Data & Infrastructure	Supabase PostgreSQL, pgvector, Edge Caching	32-dim vector cosine similarity search, relational profiles/tickets, row-level security, offline sync guarantees.

1.3 Algorithmic Cold-Start Math

For new users ($I_u = \emptyset$), Tripzy computes a synthetic lifestyle context vector from active signals: $C_u = \sum_{s \in S_u} w_s \cdot \text{vec}\{E\}(s)$. The final ranking blends lifestyle projection and SVD++ ratings via an adaptive parameter: $\text{Score} = (1 - \alpha) \cdot \text{LifestyleProjection} + \alpha \cdot \text{SVD++}$, scaling smoothly from $\alpha=0$ (pure cold start) to $\alpha=0.80$ (warm users).

2. CONSUMER EXPERIENCE & THE VIRAL FLASH LOTTERY

Feature Component	User Micro-Action	Platform Value & Signal Captured
Instagram Stories Bar	5-second auto-advancing slides with glowing gradient rings.	Dwell-time tracking feeds category interest vector; direct 'Claim Deal' CTA.
Double-Tap Heart Burst	Double-tapping deal card spawns particle hearts + haptics.	Wishlist save, +10 XP reward, instant category affinity boost in latent space.
Daily Exploration Streak	Consecutive login tracking (■ 3 Günlük Seri!).	Awards daily XP & coins, unlocking interactive mystery scratch cards.
Interactive Scratch Cards	HTML5 Canvas scratch-off foil revealing hidden discounts.	Tests price sensitivity and delivers impulse discount conversions at 45% clear.
Digital Travel Passport	City stamps (Istanbul, Cappadocia, Antalya, Bodrum).	Gamifies travel tier progression (*Çaylak Gezgin* → *Seyahat Efsanesi*).
Flash Lottery Engine	Share 9:16 Story tagging @tripzydeal or watch video ad.	Mints free lottery tickets; SHA-256 provably fair winner draw; non-winners get 25% off.

3. MASTER BUSINESS MODEL & 7 REVENUE PILLARS

3.1 The Seven Diversified Revenue Engines

Pillar	Revenue Stream	Mechanism & Economics	Target (100k MAU)
1	Booking Commissions	8% - 15% marketplace commission on completed hotel stays, dining vouchers, yacht charters.	■320,000 / mo
2	Consolation Vouchers	Automated 25% off 48-hour flash vouchers given to all non-winners (3% conversion rate).	■190,000 / mo
3	Google Rewarded Ads	Opt-in video ads (+1 ticket). Turkish eCPM: \$2.50-\$6.00; Tourist/Expat eCPM: \$15-\$35.	■300,000 / mo
4	Tripzy VIP (MRR)	■99/month or ■799/year. 3x-5x lottery odds, 100% ad-free experience, zero booking fees.	■150,000 / mo
5	Merchant Featured Drops	Phase 1: Free barter → Phase 3: ■2,500-■5,000 upfront fee for guaranteed 5k+ story shares.	■95,000 / mo
6	Owned Media Channels	Sponsored deal drops and affiliate links across Telegram, WhatsApp, Instagram (@tripzydeal).	■90,000 / mo
7	Creator Rev-Share	Influencer giveaway co-branding with performance rev-share on referred bookings.	Performance

3.2 Financial Projections Across Growth Stages

Financial Metric	10,000 MAU (Launch)	50,000 MAU (Traction)	100,000 MAU (Scale)
Total Projected Monthly Revenue	■107,500	■445,000	■1,145,000 (~\$34k/mo)
Estimated Fixed Cloud/DB Cost	~■8,500 (\$250)	~■11,500 (\$340)	~■13,500 (\$400)
Net Operating Margin %	~92.0%	~97.4%	~98.8%

4. OPERATIONAL GOVERNANCE & REGULATORY COMPLIANCE

- **Admin Panel 360° Control (/admin):** Complete management over deals, categories, users, subscriptions, 9:16 stories, flash lotteries, social raffles, fraud signals, and ticket audits.
- **Milli Piyango ■daresi (MP■) Exemption:** Lotteries operate strictly as free commercial loyalty promotions (*Promosyon ve Ticari Sadakat Kampanyas■*), exempt from gambling licensing taxes.
- **KVKK Compliance:** User notification and location tracking consents are stored with cryptographic audit timestamps in Supabase.